Recruiting and retaining drivers is seen by industry watchers as a tough battle for Ola.

Churn among drivers is high and it's very easy for drivers to stop working for the service on the fly or jump to Uber depending on the rates.

As the companies get bigger, the high churn could become a bigger problem. To find new drivers, Ola is casting a wide net, including people who don't have cars for jobs. But this acquisition is really costly.

Losing drivers frequently impacts the morale of the organization and acquiring new drivers is more expensive than retaining existing ones.

You are working as a data scientist with the Analytics Department of Ola, focused on driver team attrition. You are provided with the monthly information for a segment of drivers for 2019 and 2020 and tasked to predict whether a driver will be leaving the company or not based on their attributes like:

Why?

Hiring new drivers = 💸 expensive

Losing drivers = 📉 bad for business

Predict whether a driver will leave (attrition) or not"

👉 Isko ML language me:

Binary Classification Problem 0 = Stay 1 = Leave

overview

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
df= pd.read_csv("/content/drive/MyDrive/Aadisingh_B-41/ola_driver.csv")
df.head()
```

```
/usr/local/lib/python3.12/dist-packages/google/colab/_dataframe_summarizer.py:88: UserWarning: Cou
  cast_date_col = pd.to_datetime(column, errors="coerce")
```

	Unnamed: 0	MMM-YY	Driver_ID	Age	Gender	City	Education_Level	Income	Dateofjoining	LastWo
0	0	01/01/19	1	28.0	0.0	C23	2	57387	24/12/18	
1	1	02/01/19	1	28.0	0.0	C23	2	57387	24/12/18	
2	2	03/01/19	1	28.0	0.0	C23	2	57387	24/12/18	
3	3	11/01/20	2	31.0	0.0	C7	2	67016	11/06/20	
4	4	12/01/20	2	31.0	0.0	C7	2	67016	11/06/20	

Next steps: Generate code with df New interactive sheet

```
df.tail()
```

	Unnamed: 0	MMM-YY	Driver_ID	Age	Gender	City	Education_Level	Income	Dateofjoining	La
19099	19099	08/01/20	2788	30.0	0.0	C27	2	70254	06/08/20	
19100	19100	09/01/20	2788	30.0	0.0	C27	2	70254	06/08/20	
19101	19101	10/01/20	2788	30.0	0.0	C27	2	70254	06/08/20	
19102	19102	11/01/20	2788	30.0	0.0	C27	2	70254	06/08/20	
19103	19103	12/01/20	2788	30.0	0.0	C27	2	70254	06/08/20	

```
df.shape
```

```
(19104, 14)
```

```
df.size
```

```
267456
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19104 entries, 0 to 19103
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Unnamed: 0            19104 non-null  int64
 1   MMM-YY                19104 non-null  object
 2   Driver_ID             19104 non-null  int64
 3   Age                   19043 non-null  float64
 4   Gender                19052 non-null  float64
 5   City                  19104 non-null  object
 6   Education_Level       19104 non-null  int64
 7   Income                19104 non-null  int64
 8   Dateofjoining         19104 non-null  object
 9   LastWorkingDate       1616 non-null   object
 10  Joining Designation   19104 non-null  int64
```

```
 11  Grade                 19104 non-null  int64
 12  Total Business Value  19104 non-null  int64
 13  Quarterly Rating      19104 non-null  int64
dtypes: float64(2), int64(8), object(4)
memory usage: 2.0+ MB
```

```
df.describe()
```

	Unnamed: 0	Driver_ID	Age	Gender	Education_Level	Income	Desi
count	19104.000000	19104.000000	19043.000000	19052.000000	19104.000000	19104.000000	19104
mean	9551.500000	1415.591133	34.668435	0.418749	1.021671	65652.025126	1
std	5514.994107	810.705321	6.257912	0.493367	0.800167	30914.515344	0
min	0.000000	1.000000	21.000000	0.000000	0.000000	10747.000000	1
25%	4775.750000	710.000000	30.000000	0.000000	0.000000	42383.000000	1
50%	9551.500000	1417.000000	34.000000	0.000000	1.000000	60087.000000	1
75%	14327.250000	2137.000000	39.000000	1.000000	2.000000	83969.000000	2
max	19103.000000	2788.000000	58.000000	1.000000	2.000000	188418.000000	5

```
df.isnull().sum()
```

	0
Unnamed: 0	0
MMM-YY	0
Driver_ID	0
Age	61
Gender	52
City	0
Education_Level	0
Income	0
Dateofjoining	0
LastWorkingDate	17488
Joining Designation	0
Grade	0
Total Business Value	0
Quarterly Rating	0

dtype: int64

A small portion of records is missing age (61 cases) and gender (52 cases), which is minimal and can be reasonably estimated without affecting analysis.

A large number of records (17,488) are missing last working date, but this is expected.

```
df['Age'] = df['Age'].fillna(df['Age'].median())
```

```
df['Gender'].value_counts()
```

Gender	count
0.0	11074
1.0	7978

dtype: int64

Assume 0 denotes for men and 1 for women

```
df['Gender']=df['Gender'].fillna(df['Gender'].mode()[0])
```

If last working date is missing → driver is still working

change null value to current data

```
df.drop('Unnamed: 0', axis=1, inplace=True)
```

```
df.isnull().sum()
```

	0
MMM-YY	0
Driver_ID	0
Age	0
Gender	0
City	0
Education_Level	0
Income	0
Dateofjoining	0
LastWorkingDate	17488
Joining Designation	0
Grade	0
Total Business Value	0
Quarterly Rating	0

dtype: int64

Define target (label)

Churn label per row (driver-month): churn = 1 if LastWorkingDate exists and LastWorkingDate ≤ reporting month end; else 0.

descriptive Analysis

```
df.describe()
```

	Driver_ID	Age	Gender	Education_Level	Income	Joining Designation	
count	19104.000000	19104.000000	19104.000000	19104.000000	19104.000000	19104.000000	19104
mean	1415.591133	34.666300	0.417609	1.021671	65652.025126	1.690536	2
std	810.705321	6.248026	0.493178	0.800167	30914.515344	0.836984	1
min	1.000000	21.000000	0.000000	0.000000	10747.000000	1.000000	1
25%	710.000000	30.000000	0.000000	0.000000	42383.000000	1.000000	1
50%	1417.000000	34.000000	0.000000	1.000000	60087.000000	1.000000	2
75%	2137.000000	39.000000	1.000000	2.000000	83969.000000	2.000000	3
max	2788.000000	58.000000	1.000000	2.000000	188418.000000	5.000000	5

lets check how many drivers are exists

```
df['Driver_ID'].value_counts()
```

Driver_ID	count
2784	24
2753	24
2771	24
1745	24
2766	24
...	...
136	1
138	1
2710	1
1744	1
2036	1

2381 rows × 1 columns

dtype: int64

Drivers with higher counts are more active and contributing more to operations

count of unique drivers !

```
unique_drivers=df['Driver_ID'].unique()
print(len(unique_drivers))
```

```
2381
```

```
df['Driver_ID'].value_counts().count()
```

```
np.int64(2381)
```

👉 value_counts() gives frequency

👉 .count() gives number of unique IDs

temporal Analysis

How many drivers joined and left each month?

```
df['Dateofjoining'].head()
```

	Dateofjoining
0	24/12/18
1	24/12/18
2	24/12/18
3	11/06/20
4	11/06/20

dtype: object

```
print(df['LastWorkingDate'].dtype)
```

```
object
```

```
df['Dateofjoining'] = pd.to_datetime(df['Dateofjoining'],format='%d-%m-%Y', errors='coerce')
df['LastWorkingDate'] = pd.to_datetime(df['LastWorkingDate'],format='%d-%m-%Y', errors='coerce')
```

```
print(df.columns)
print(df.dtypes)
```

```
Index(['MMM-YY', 'Driver_ID', 'Age', 'Gender', 'City', 'Education_Level',
       'Income', 'Dateofjoining', 'LastWorkingDate', 'Joining Designation',
       'Grade', 'Total Business Value', 'Quarterly Rating'],
      dtype='object')
MMM-YY                         object
Driver_ID                       int64
Age                           float64
Gender                        float64
City                           object
Education_Level                 int64
Income                          int64
Dateofjoining           datetime64[ns]
LastWorkingDate         datetime64[ns]
Joining Designation             int64
Grade                           int64
Total Business Value            int64
Quarterly Rating                int64
dtype: object
```

why dont use count here ?

- 3. .fillna(0) Replaces missing values (NaN) with 0

we determine the average tenure of drivers in the dataset?

feature Engineering

Feature engineering means:

Creating, modifying, or selecting variables (features) from raw data so that models can learn better patterns.

- Why is it Important?

Improves model accuracy 📈 Helps models learn faster ⚡ Reduces overfitting Makes data more meaningful

```
df=pd.read_csv("/content/drive/MyDrive/Aadisingh_B-41/ola_driver.csv")
df.head()
```

```
/usr/local/lib/python3.12/dist-packages/google/colab/_dataframe_summarizer.py:88: UserWarning: Cou
  cast_date_col = pd.to_datetime(column, errors="coerce")
```

	Unnamed: 0	MMM-YY	Driver_ID	Age	Gender	City	Education_Level	Income	Dateofjoining	LastWo
0	0	01/01/19	1	28.0	0.0	C23	2	57387	24/12/18	
1	1	02/01/19	1	28.0	0.0	C23	2	57387	24/12/18	
2	2	03/01/19	1	28.0	0.0	C23	2	57387	24/12/18	
3	3	11/01/20	2	31.0	0.0	C7	2	67016	11/06/20	
4	4	12/01/20	2	31.0	0.0	C7	2	67016	11/06/20	

Next steps: Generate code with df New interactive sheet

Target variable is LastWorkingDays

now a question is how many drivers are still working and how many are left !

create a column name still_drivers with assumtion

1-> left 0-> active

shows in df['churn'] column

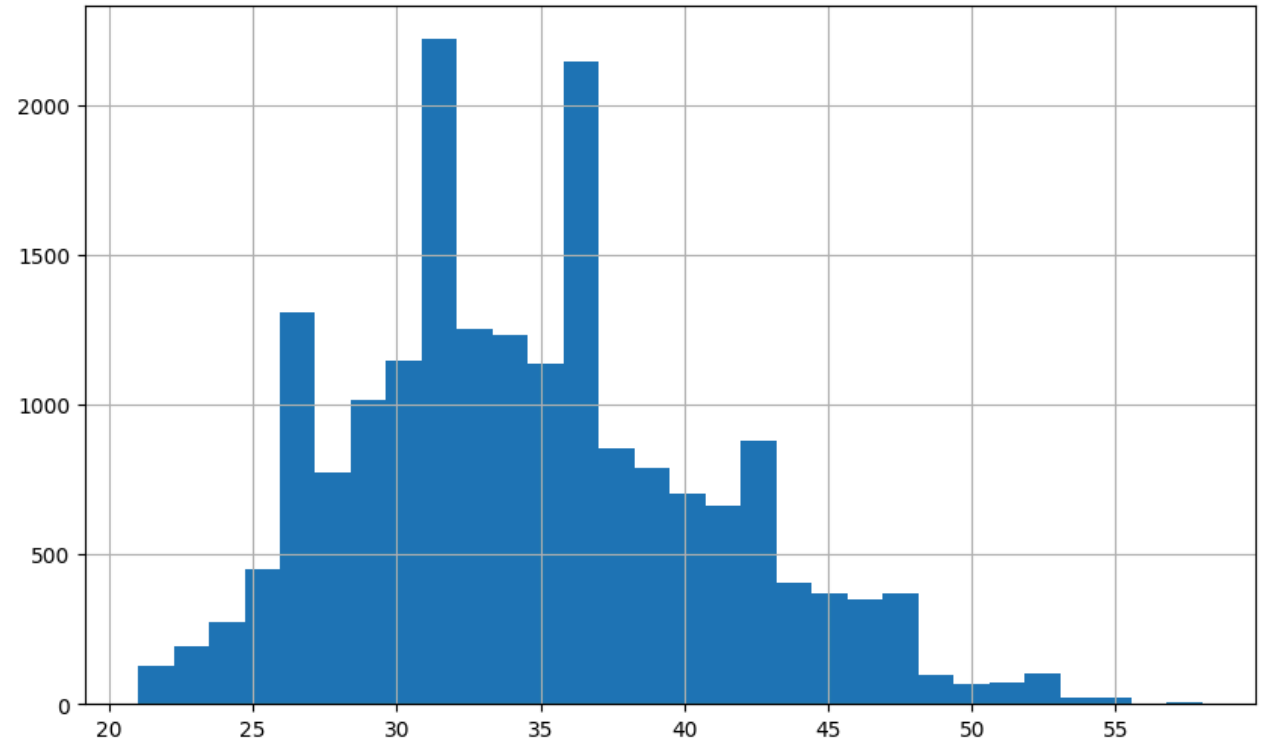
Eda

```
# Summary stats
print(df[['Age', 'Income', 'Total Business Value']].describe())
```

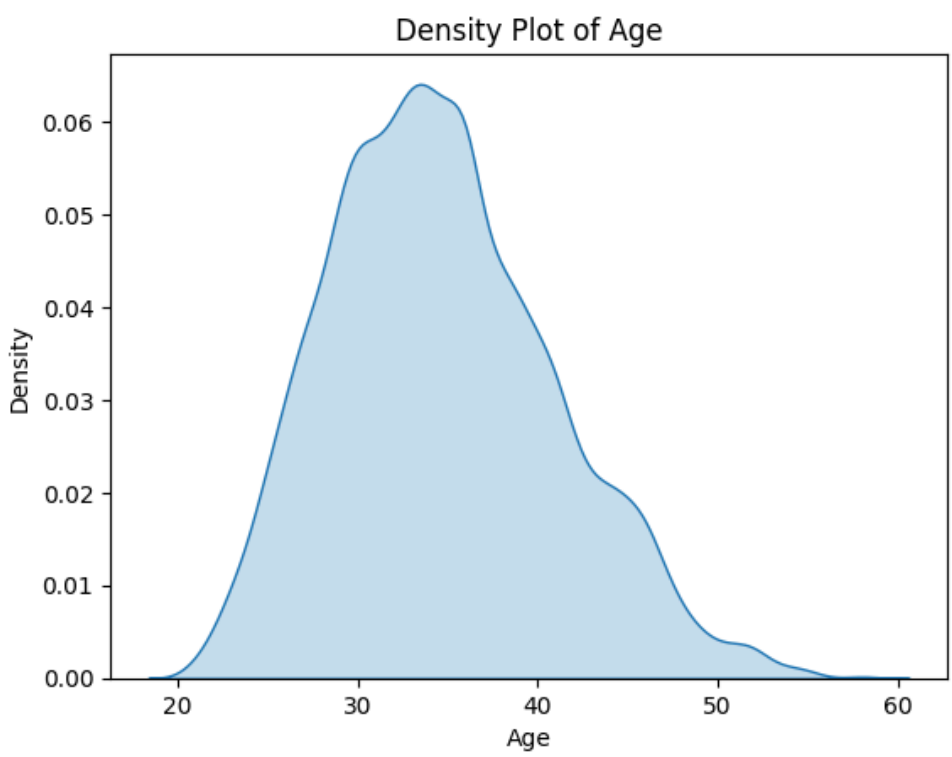
```
                Age         Income  Total Business Value
count  19043.000000   19104.000000          1.910400e+04
mean      34.668435   65652.025126          5.716621e+05
std        6.257912   30914.515344          1.128312e+06
min       21.000000   10747.000000         -6.000000e+06
25%       30.000000   42383.000000          0.000000e+00
50%       34.000000   60087.000000          2.500000e+05
75%       39.000000   83969.000000          6.997000e+05
max       58.000000  188418.000000          3.374772e+07
```

```
df['Age'].hist(bins=30,figsize=(10,6))
```

```
<Axes: >
```




```
sns.kdeplot(df['Age'], fill=True)

plt.title("Density Plot of Age")
plt.xlabel("Age")
plt.ylabel("Density")
plt.show()
```



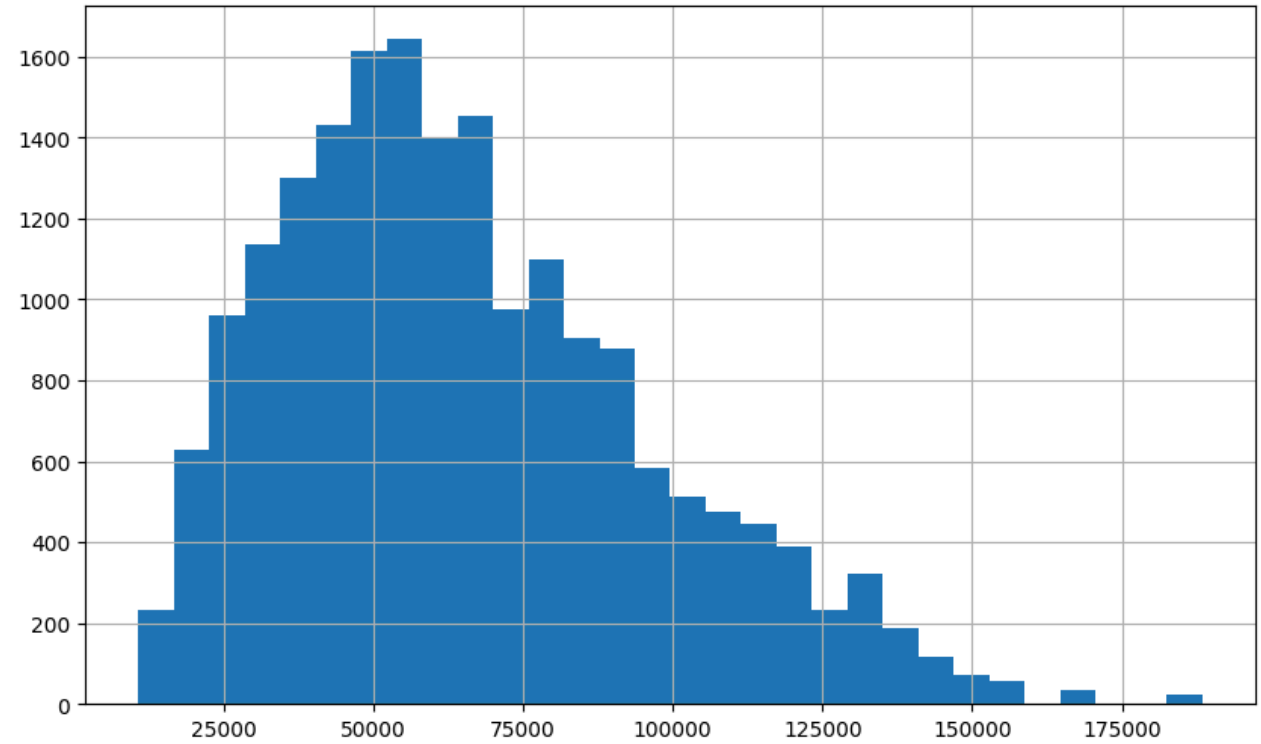
The highest peaks are observed at 32 and 37 years.

This suggests that the dataset is dominated by mid-age working individuals, especially those in their early to mid-30s.

Since the distribution is slightly skewed, it indicates that: There are fewer people in the younger and older age groups compared to the middle-age group.

```
df['Income'].hist(bins=30,figsize=(10,6))
```

```
<Axes: >
```



The majority of incomes lie between ₹40,000 to ₹65,000, indicating that most individuals fall in the mid-income group.

The distribution has a long tail on the right side, which shows the presence of few high-income individuals.

to detect outliers

```
df['Income'].describe()
```

	Income
count	19104.000000
mean	65652.025126
std	30914.515344
min	10747.000000
25%	42383.000000
50%	60087.000000
75%	83969.000000
max	188418.000000

dtype: float64

IQR

```
in_25=np.percentile(df['Income'],25)
print(in_25)
```

```
42383.0
```

“Around 25% of the customers earn ₹42,383 or less. This represents the lower-income segment of the population, while the remaining 75% earn above this level

```
in_50=np.percentile(df['Income'],50)
print(in_50)
```

```
60087.0
```

“Around 50% of the customers earn ₹60,087 or less.

```
in_75=np.percentile(df['Income'],75)
print(in_75)
```

```
83969.0
```

“Around 75% of the customers earn ₹83,969 or less.

```
iqr_income=in_75-in_25
print(iqr_income)
```

```
41586.0
```

```
normal_range=(df['Income'].max() - df['Income'].min())
print(normal_range)
```

```
177671
```

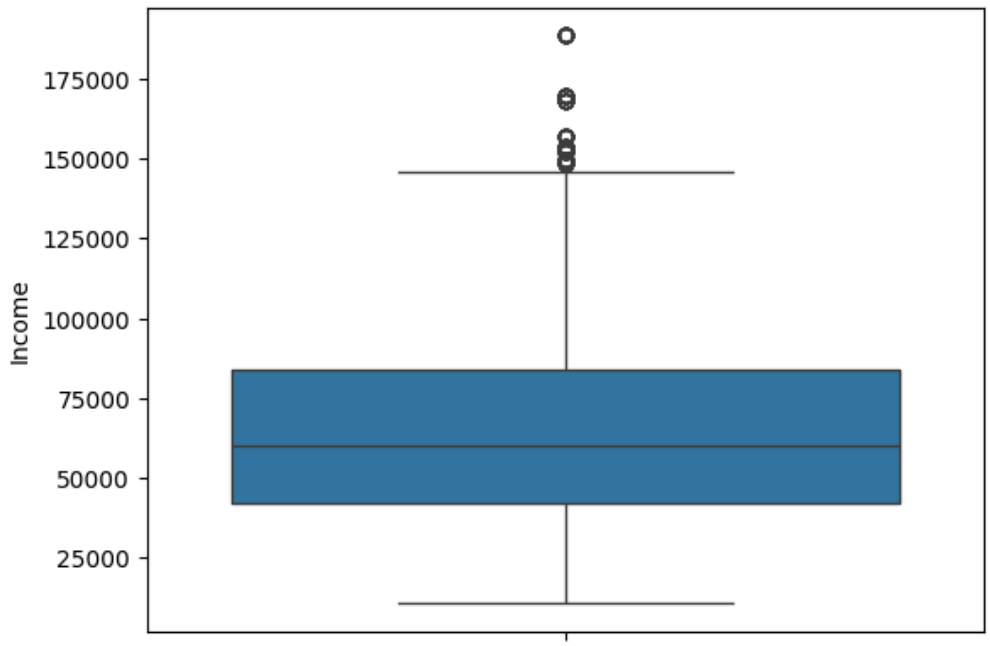
The Interquartile Range (IQR) of income is ₹41,586.

This means that the middle 50% of the population's income lies within a range of ₹41,586.

The normal range of income (Maximum – Minimum) is ₹177,671.

This shows that the overall income distribution is highly spread out, covering both very low and very high-income individuals.

```
sns.boxplot(df['Income'])
plt.show()
```




```
 # upper limit = Q3 + 1.5 * IQR

upper = 83969 + 1.5*(iqr_income)
print(upper)
```

```
146348.0
```

```
# all the values greater than upper bound are outliers
outliers_income = df[df["Income"] > upper]

# number of outliers
len(outliers_income)
```

```
188
```

```
(188/19104)*100
```

```
0.9840871021775545
```

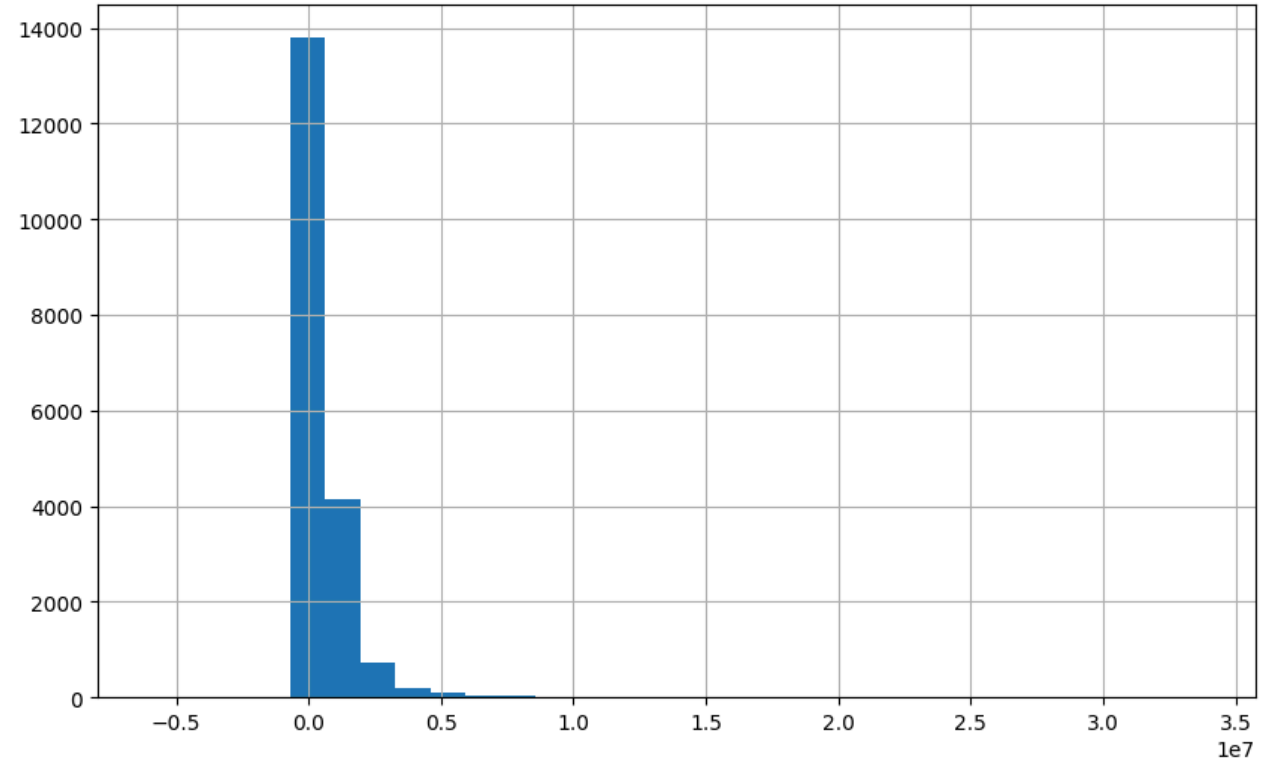
this indicates that less than 1% of the population behaves significantly differently from the majority.

- Key business insights: The dataset is highly stable and mostly consistent Only a very small segment of customers shows extreme behavior These extreme values may represent: High-value premium customers (in income case) Or unusual/rare cases in the business

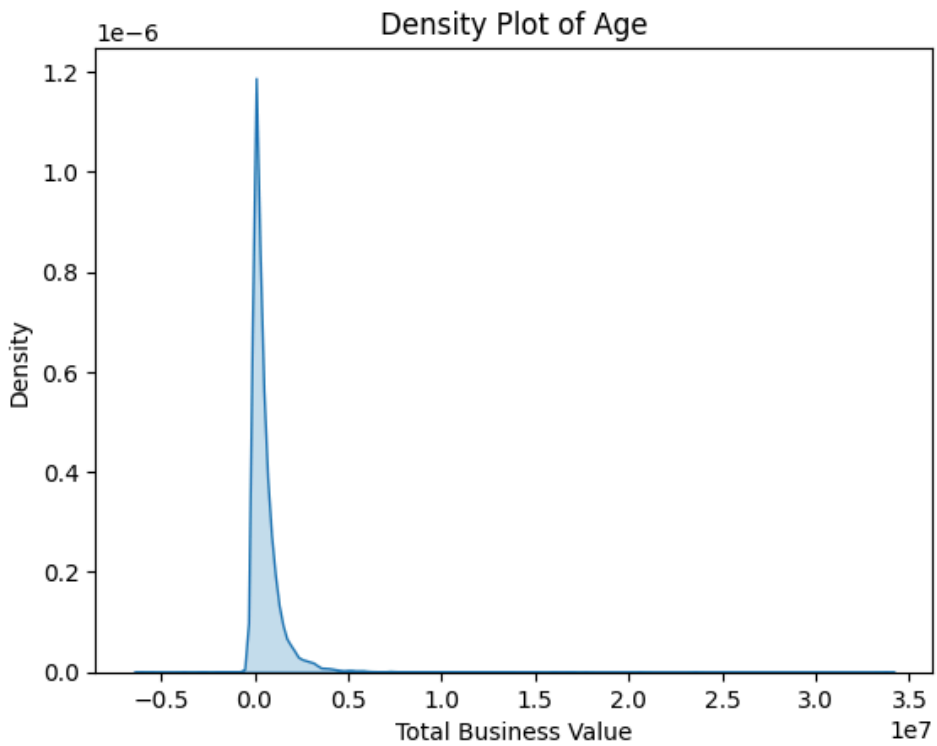
These outliers represent premium/high-value customers

```
df['Total Business Value'].hist(bins=30,figsize=(10,6))
```

```
<Axes: >
```




```
sns.kdeplot(df['Total Business Value'], fill=True)

plt.title("Density Plot of Age")
plt.xlabel("Total Business Value")
plt.ylabel("Density")
plt.show()
```



Heavy skew (most values near 0) 📌 Very high positive + some negative value

use IQR Technique for total business value

```
Total_25=np.percentile(df['Total Business Value'],25)
print(Total_25)
```

```
0.0
```

25% of your employees records have:

Zero Total Business Value

```
Total_50=np.percentile(df['Total Business Value'],50)
print(Total_50)
```

```
250000.0
```

50% of your data points generate ≤ 250,000 value This represents your “middle-performing segment”

```
Total_75=np.percentile(df['Total Business Value'],75)
print(Total_75)
```

```
699700.0
```

Top 25% of records generate more than 699,700 value These are your high-value contributors

```
iqr_totalBusinessValue=Total_75-Total_25
print(iqr_totalBusinessValue)
```

```
699700.0
```

```
normal_rangeOftotal=(df['Total Business Value'].max() - df['Total Business Value'].min())
print(normal_range)
```

```
177671
```

There is huge variability in performance Some generate almost nothing (Q1 = 0) Some generate very high value (Q3 ~ 700K)

👉 Conclusion:

business is highly uneven and skewed — not stable or uniform

```
# df['Total Business Value'].max()
df['Total Business Value'].min()
```

```
-6000000
```

range>=Q3

```
upper_total=699700+1.5*(iqr_totalBusinessValue)
print(upper_total)
```

```
1749250.0
```

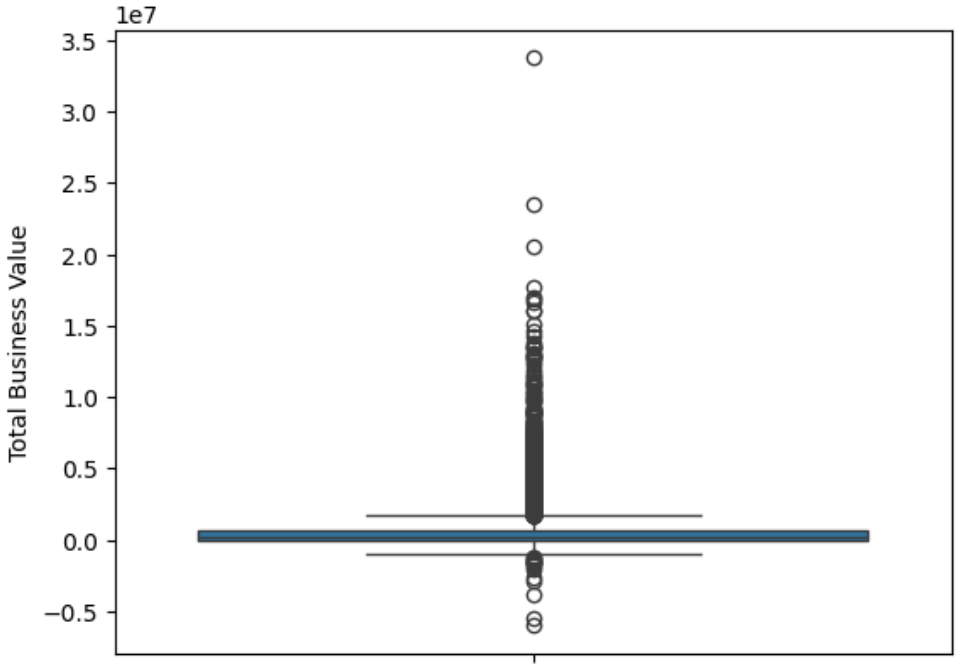
```
outliers_TotalBusinessValue = df[df["Total Business Value"] > upper_total]

# number of outliers
len(outliers_TotalBusinessValue)
```

```
1355
```

```
(1355/19104)*100
```

```
7.092755443886096
```

```
sns.boxplot(df['Total Business Value'])
plt.show()
```



The Total Business Value distribution is extremely skewed, with most observations clustered near zero and a few extreme high-value contributors. This indicates a strong dependency on a small group of high performers, following a Pareto-like distribution. The presence of negative values suggests possible losses, refunds, or data inconsistencies that require validation. Overall, the business is not failing, but highly concentrated and vulnerable to top-performer dependency, making segmentation and retention strategies critica

How does Quarterly Rating vary across different drivers and time periods?

```
drivers_rating=df['Driver_ID'].groupby(df['Quarterly Rating']).count().reset_index()
print(drivers_rating)
```

```
   Quarterly Rating  Driver_ID
0                 1       7679
1                 2       5553
2                 3       3895
3                 4       1977
```

```
df['Dateofjoining'] = pd.to_datetime(df['Dateofjoining'], format='%d/%m/%y')
df['Quarter']=df['Dateofjoining'].dt.to_period('Q')
print(df['Quarter'])
```

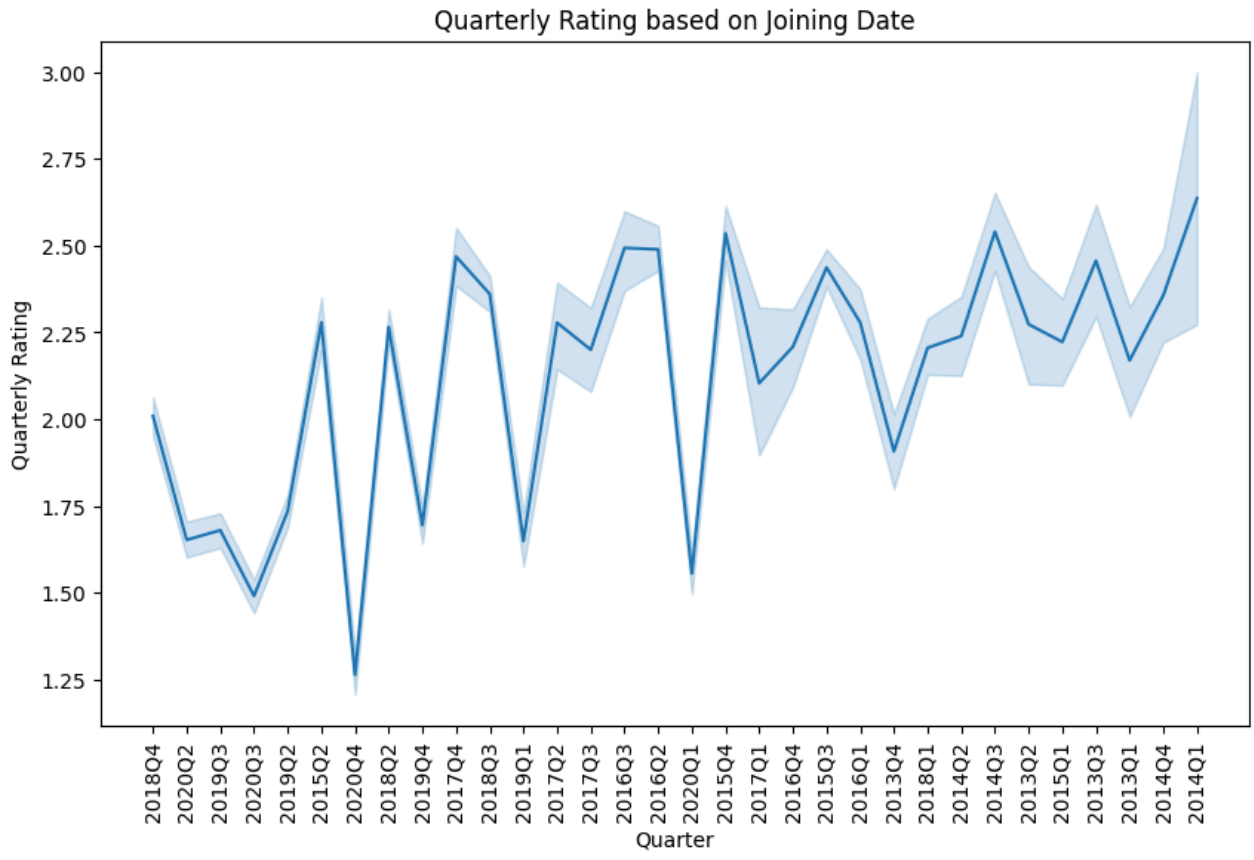
```
0        2018Q4
1        2018Q4
2        2018Q4
3        2020Q2
4        2020Q2
          ...
19099    2020Q3
19100    2020Q3
19101    2020Q3
19102    2020Q3
19103    2020Q3
Name: Quarter, Length: 19104, dtype: period[Q-DEC]
```

Q1 → Jan, Feb, Mar Q2 → Apr, May, Jun Q3 → Jul, Aug, Sep Q4 → Oct, Nov, Dec

```
quarter_rating = df.groupby('Quarter')['Quarterly Rating'].mean().reset_index()
print(quarter_rating)
df['Quarter'] = df['Quarter'].astype(str)
```

```
   Quarter  Quarterly Rating
0   2013Q1          2.169118
1   2013Q2          2.273381
2   2013Q3          2.456376
3   2013Q4          1.907063
4   2014Q1          2.636364
5   2014Q2          2.239130
6   2014Q3          2.540107
7   2014Q4          2.356784
8   2015Q1          2.222222
9   2015Q2          2.278618
10  2015Q3          2.436722
11  2015Q4          2.535533
12  2016Q1          2.278515
13  2016Q2          2.489117
14  2016Q3          2.493392
15  2016Q4          2.208333
16  2017Q1          2.103448
17  2017Q2          2.277778
18  2017Q3          2.200000
19  2017Q4          2.468582
20  2018Q1          2.205318
21  2018Q2          2.265636
22  2018Q3          2.360502
23  2018Q4          2.008864
24  2019Q1          1.648496
25  2019Q2          1.734534
26  2019Q3          1.679619
27  2019Q4          1.694486
28  2020Q1          1.554737
29  2020Q2          1.651888
30  2020Q3          1.490157
31  2020Q4          1.262940
```

```
plt.figure(figsize=(10,6))
sns.lineplot(data=df, x='Quarter', y='Quarterly Rating')
plt.xticks(rotation=90)
plt.title("Quarterly Rating based on Joining Date")
plt.show()
```



o Are there any trends or patterns in the monthly income or business value acquired?

```
df['Income'].describe()
```

	Income
count	19104.000000
mean	65652.025126
std	30914.515344
min	10747.000000
25%	42383.000000
50%	60087.000000
75%	83969.000000
max	188418.000000

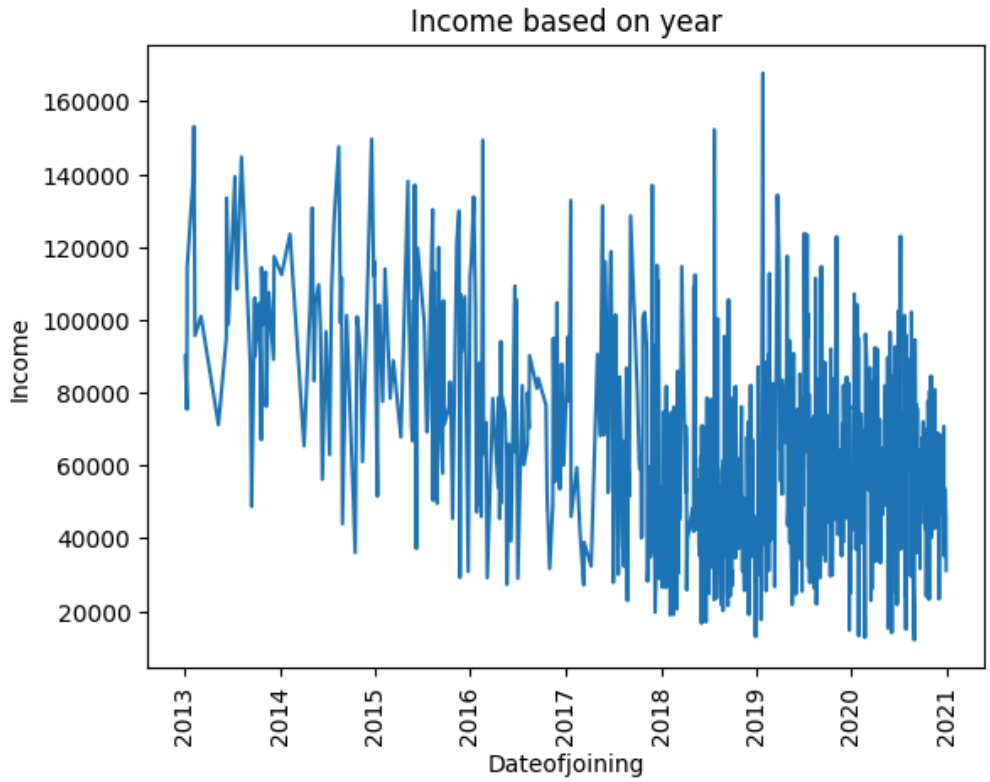
dtype: float64

Assuming Income represents monthly earnings based on realistic salary ranges...

The income distribution is right-skewed, indicating that most individuals earn moderate monthly income, while a small number of individuals earn significantly higher amounts.

```
Income=df['Income'].groupby(df['Dateofjoining']).mean()
```

```
sns.lineplot(data=Income)
plt.xticks(rotation=90)
plt.title("Income based on year")
plt.show()
```



The average yearly income shows a gradual declining trend over time.

```
df['Join_Year'] = pd.to_datetime(df['Dateofjoining']).dt.year

df['Income'].groupby(df['Join_Year']).mean()
```

Join_Year	Income
2013	104466.467532
2014	96026.733002
2015	84773.602036
2016	71145.622769
2017	66612.708182
2018	54937.635130
2019	60659.958804
2020	60921.538315

dtype: float64

Older joiners earning more:

"Drivers who joined earlier tend to have higher income, possibly due to experience or tenure benefits."

New joiners earning less:

"Recently joined drivers have lower income, indicating a ramp-up period."

```
df['Dateofjoining'] = pd.to_datetime(df['Dateofjoining'], errors='coerce').dt.normalize()
df['LastWorkingDate'] = pd.to_datetime(df['LastWorkingDate'], errors='coerce').dt.normalize()

joins = df.groupby(df['Dateofjoining'].dt.to_period('M')).size()
leave = df.groupby(df['LastWorkingDate'].dt.to_period('M')).size()

# Align the series to ensure that months with no joins/leaves are handled correctly
all_months = pd.period_range(start=min(joins.index.min(), leave.index.min()),
                             end=max(joins.index.max(), leave.index.max()), freq='M')
joins = joins.reindex(all_months, fill_value=0)
leave = leave.reindex(all_months, fill_value=0)

summary = pd.DataFrame({
    'Joined': joins,
    'Left': leave
}).fillna(0)

summary.head()
```

```
/tmp/ipykernel_4773/1324767914.py:2: UserWarning: Could not infer format, so each element will be
  df['LastWorkingDate'] = pd.to_datetime(df['LastWorkingDate'], errors='coerce').dt.normalize()
```

	Joined	Left
2013-01	62	0
2013-02	50	0
2013-03	24	0
2013-04	0	0
2013-05	32	0

Next steps: Generate code with summary New interactive sheet

```
summary['year']=summary.index.to_timestamp().year
```

```
yearly_summary = summary.groupby('year').sum()

print(yearly_summary)
```

```
      Joined  Left
year
2013     693     0
2014     603     0
2015    1965     0
2016    1625     0
2017    1100     0
2018    4936     5
2019    4515   825
2020    3667   786
```

Higher churn is observed among recently joined drivers...

Statistical test (strong proof) later

```
df['Total Business Value'].describe()
```

	Total Business Value
count	1.910400e+04
mean	5.716621e+05
std	1.128312e+06
min	-6.000000e+06
25%	0.000000e+00
50%	2.500000e+05
75%	6.997000e+05
max	3.374772e+07

dtype: float64

```
df.groupby('Join_Year')['Total Business Value'].mean()
```

Join_Year	Total Business Value
2013	1.213787e+06
2014	1.128657e+06
2015	8.864929e+05
2016	7.855546e+05
2017	7.454556e+05
2018	5.636233e+05
2019	4.176330e+05
2020	2.435653e+05

dtype: float64

In the initial years, the organization had a smaller workforce, resulting in lower total business revenue.

As hiring increased over time, the growing workforce contributed to a steady rise in total business value.

The business reached its peak performance around 2015–2018, coinciding with the highest level of employee hiring and workforce expansion. After this peak period, a gradual increase in employee attrition was observed, particularly among newly hired drivers.

The rising attrition negatively impacted workforce stability and reduced overall productivity.

As a result, the total business value began to decline in the later years, indicating potential challenges in employee retention and operational efficiency.

How should missing values in LastWorkingDate be treated, considering it indicates whether a driver has left?

```
df['Churn'] = df['LastWorkingDate'].notna().astype(int)
df[['Churn','LastWorkingDate']].head()
```

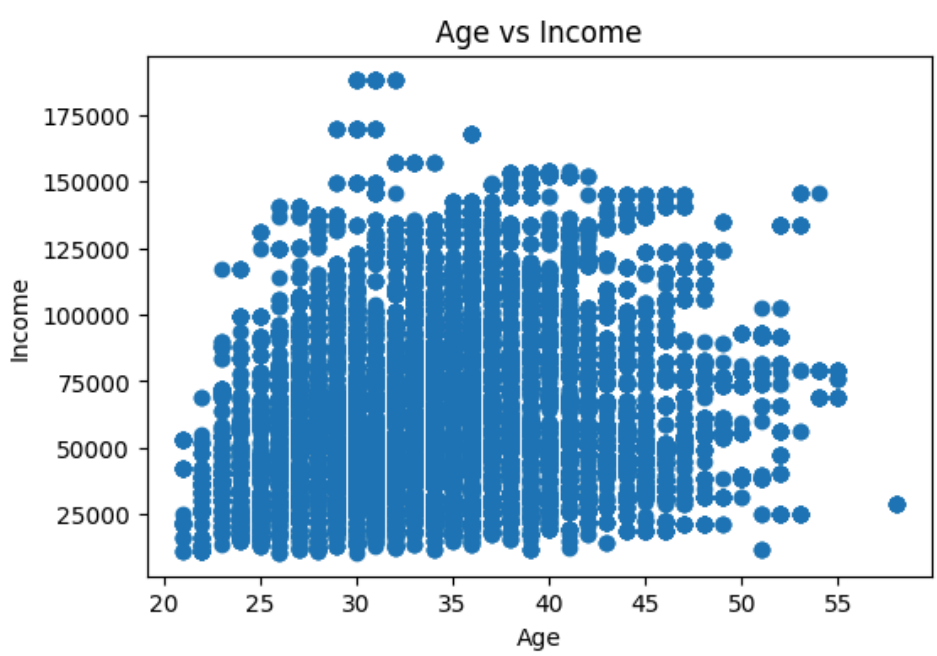
	Churn	LastWorkingDate
0	0	NaT
1	0	NaT
2	1	2019-03-11
3	0	NaT
4	0	NaT

📌 Final result 1 = Driver has left (churned) 0 = Driver is still active

the LastWorkingDate column, where values are missing, we can treat them as 'Still Working'. Since the data type is object, we are converting it into a datetime format and filling the missing values with today's date

Correlation and Relationships:

```
plt.figure(figsize=(6,4))
plt.scatter(df['Age'], df['Income'])
plt.xlabel('Age')
plt.ylabel('Income')
plt.title('Age vs Income')
plt.show()
```



👉 Conclusion: Weak positive correlation

```
corr = df['Age'].corr(df['Income'])
print(corr)
```

```
0.19111177421789227
```

No age bias in hiring

Young drivers can perform as well as experienced ones Hiring should focus on skills, not age

Income variability = performance-driven system Incentives, effort, and efficiency likely impact earnings more

How do Education_Level and City affect Total Business Value

Since both features are categorical, we DON'T use correlation directly.

```
df.groupby('Education_Level')['Total Business Value'].mean()
```

Education_Level	Total Business Value
0	565410.657872
1	601287.867133
2	545364.175755

dtype: float64

```
df['Total Business Value'].groupby(df['City']).mean().sort_values(ascending=False)
```

City	Total Business Value
C13	796263.075571
C29	736637.511111
C12	667282.310867
C26	661837.445339
C5	634855.975610
C16	632585.712271
C19	630978.151986
C14	607931.635802
C28	591406.778917
C24	584712.426710
C21	572684.776119
C27	572039.312977
C8	566328.539326
C6	566042.954545
C22	559749.431397
C4	556092.266436
C2	553365.084746
C15	553266.636005
C18	550106.250000
C10	540753.736559
C11	538549.145299
C1	531560.280650
C25	507575.119863
C7	484569.228243
C20	468535.605159
C9	467914.865385
C3	458003.940345
C17	429160.204545
C23	423986.561338

dtype: float64

```
pd.set_option('display.float_format', '{:.2f}'.format)
```

```
df.groupby(['City', 'Education_Level'])['Total Business Value'].mean().sort_values(ascending=False
```

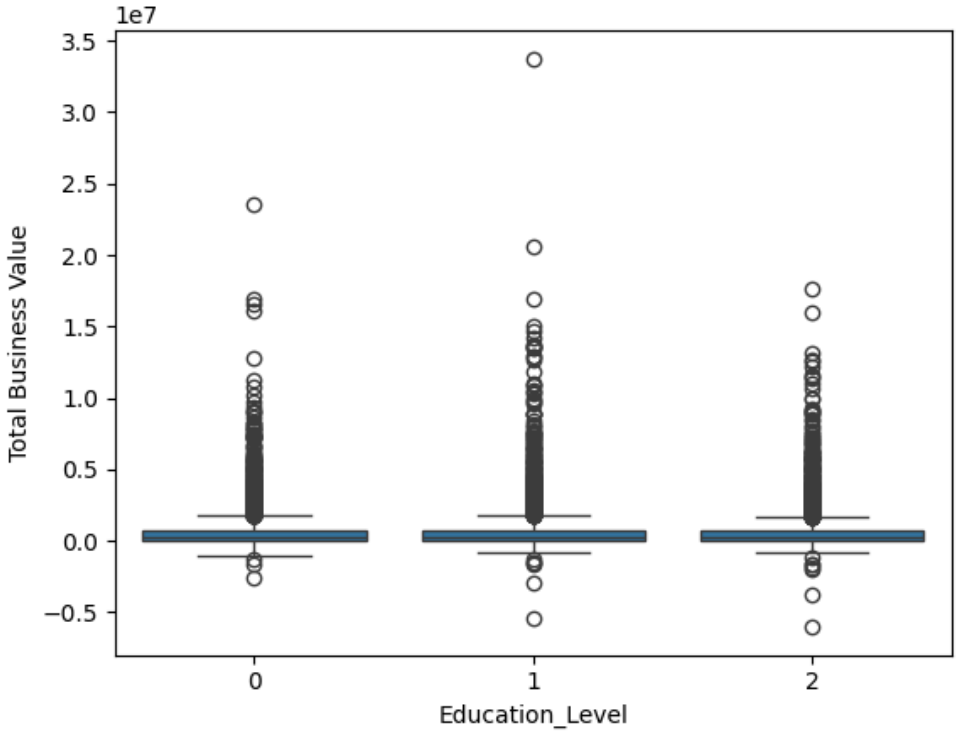
City	Education_Level	Total Business Value
C13	**1**	1037106.39
	0	907502.60
C12	**0**	865403.03
C29	**0**	854741.25
C26	**0**	784708.12
...	**...**	...
C14	**1**	231839.50
C17	**2**	225343.46
C11	**0**	206695.29
C23	**0**	201321.55
C9	**0**	158471.67

87 rows × 1 columns

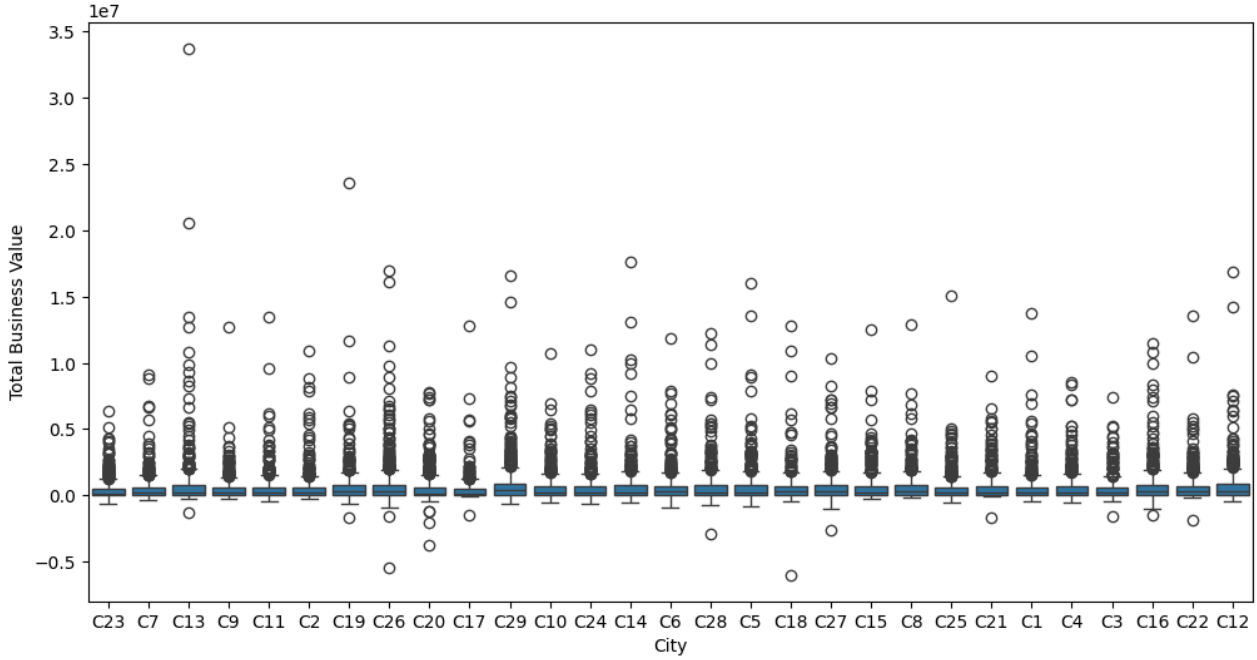
dtype: float64

Which education level performs best in each city

```
sns.boxplot(x='Education_Level', y='Total Business Value', data=df)
plt.show()
```



```
plt.figure(figsize=(12,6))
sns.boxplot(x='City', y='Total Business Value', data=df)
plt.show()
```

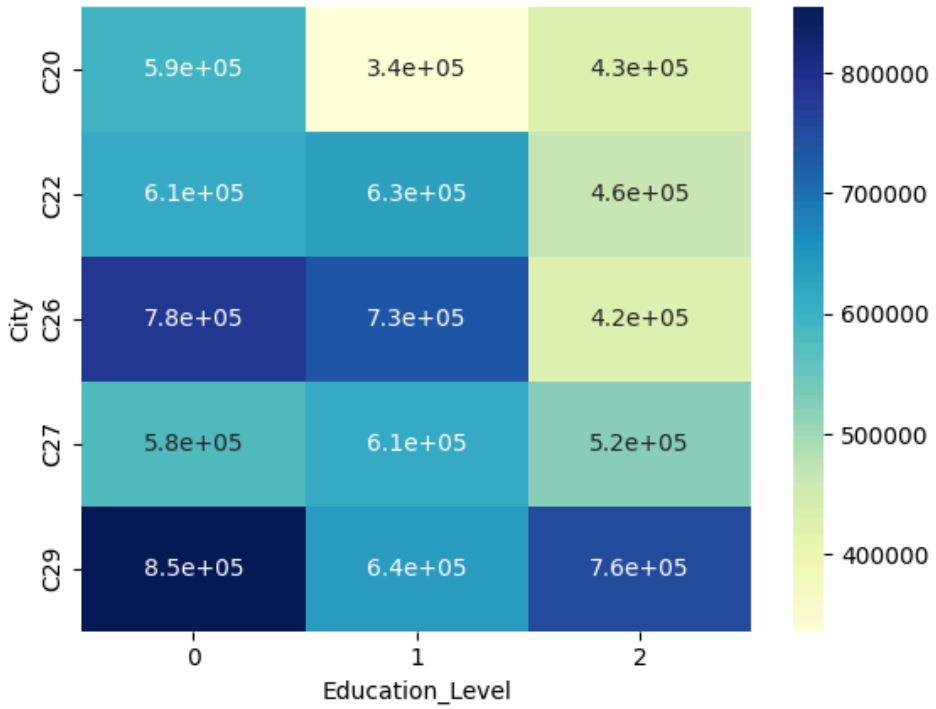


"What is the average business value for each City + Education Level combination?"

```
top_cities = df['City'].value_counts().index[:5]
df_small = df[df['City'].isin(top_cities)]
```

```
pivot = df_small.pivot_table(
    values='Total Business Value',
    index='City',
    columns='Education_Level',
    aggfunc='mean'
)

sns.heatmap(pivot, annot=True, cmap='YlGnBu')
```

```
<Axes: xlabel='Education_Level', ylabel='City'>
```



o Are drivers with higher Quarterly Rating more likely to stay longer?

```
df['LastWorkingDate']=df['LastWorkingDate'].fillna(pd.to_datetime('today'))
```

```
df['Tenure'] = (df['LastWorkingDate'] - df['Dateofjoining']).dt.days
```

```
df.groupby('Quarterly Rating')['Tenure'].mean()
```

Quarterly Rating	Tenure
1	2201.99
2	3018.80
3	3187.85
4	3025.08

dtype: float64

```
df.groupby('Quarterly Rating')['Churn'].mean()
```

Quarterly Rating	Churn
1	0.19
2	0.03
3	0.01
4	0.01

dtype: float64

```
df[['Quarterly Rating', 'Tenure']].corr()
```

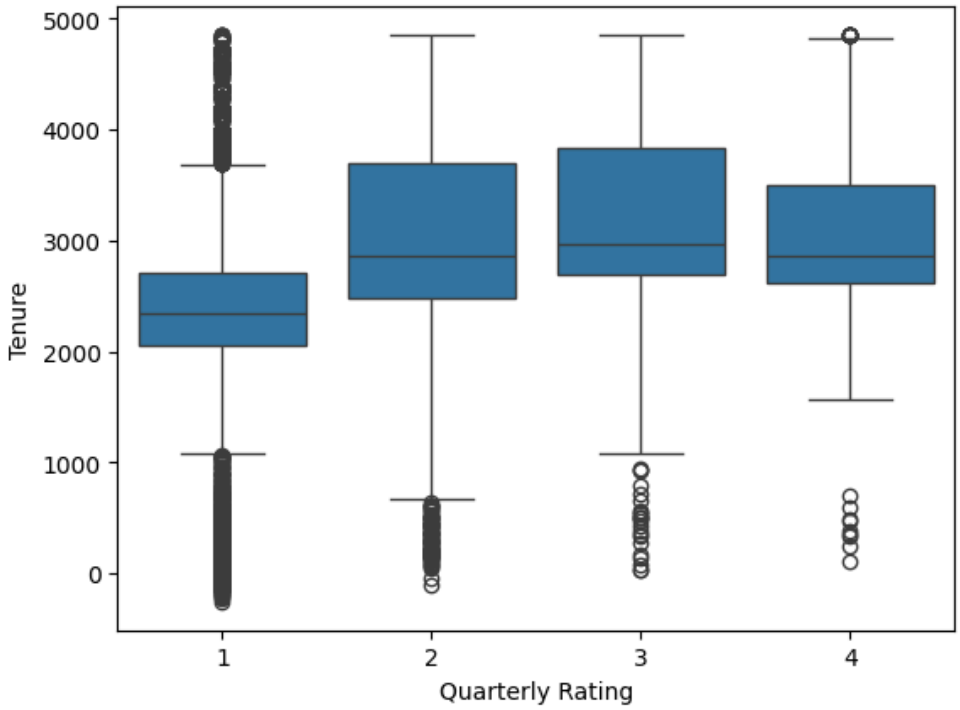
	Quarterly Rating	Tenure
Quarterly Rating	1.00	0.36
Tenure	0.36	1.00

```
df[['Quarterly Rating', 'Churn']].corr()
```

	Quarterly Rating	Churn
Quarterly Rating	1.00	-0.26
Churn	-0.26	1.00

```
sns.boxplot(x='Quarterly Rating', y='Tenure', data=df)
```

```
<Axes: xlabel='Quarterly Rating', ylabel='Tenure'>
```




```
df.head()
```

	Unnamed: 0	MMM-YY	Driver_ID	Age	Gender	City	Education_Level	Income	Dateofjoining	LastW
0	0	01/01/19	1	28.00	0.00	C23	2	57387	2018-12-24	14:5
1	1	02/01/19	1	28.00	0.00	C23	2	57387	2018-12-24	14:5
2	2	03/01/19	1	28.00	0.00	C23	2	57387	2018-12-24	00:0
3	3	11/01/20	2	31.00	0.00	C7	2	67016	2020-06-11	14:5
4	4	12/01/20	2	31.00	0.00	C7	2	67016	2020-06-11	14:5

Next steps: Generate code with df New interactive sheet

```
# Target column (sirf ek column)
y = df['Churn']

# Baaki saare columns (Churn hata ke)
X = df.drop('Churn', axis=1)
```

```
print(X.shape)
print(y.shape)
```

```
(19104, 17)
(19104,)
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

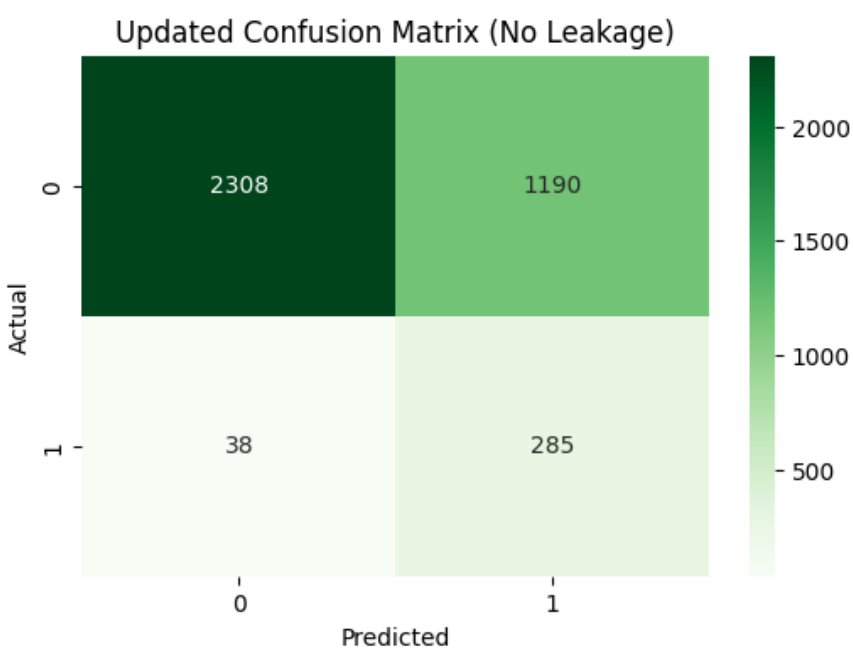
```
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer

scaler = StandardScaler()
imputer = SimpleImputer(strategy='median')

# Adding 'Tenure' to the drop list to prevent data leakage
columns_to_drop = ['Unnamed: 0', 'MMM-YY', 'City', 'Dateofjoining', 'LastWorkingDate', 'Quarter',

X_train_numeric = X_train.drop(columns=columns_to_drop, errors='ignore')
X_test_numeric = X_test.drop(columns=columns_to_drop, errors='ignore')

# Impute any remaining NaNs before scaling
X_train_imputed = imputer.fit_transform(X_train_numeric)
X_test_imputed = imputer.transform(X_test_numeric)

X_train_scaled = scaler.fit_transform(X_train_imputed)
X_test_scaled = scaler.transform(X_test_imputed)
```

```
from sklearn.linear_model import LogisticRegression

# Re-initializing and fitting the model on the updated scaled data
model = LogisticRegression(class_weight='balanced', max_iter=1000)
model.fit(X_train_scaled, y_train)
```

```
LogisticRegression
LogisticRegression(class_weight='balanced', max_iter=1000)
```

```
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix


# Re-evaluating the model after fixing data leakage
y_pred = model.predict(X_test_scaled)

print('Updated Accuracy Score:', accuracy_score(y_test, y_pred))
print('\nUpdated Classification Report:')
print(classification_report(y_test, y_pred))

# Plot Updated Confusion Matrix
plt.figure(figsize=(6,4))
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='Greens')
plt.title('Updated Confusion Matrix (No Leakage)')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```

```
Updated Accuracy Score: 0.6786181627846114

Updated Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.66      0.79      3498
           1       0.19      0.88      0.32       323

    accuracy                           0.68      3821
   macro avg       0.59      0.77      0.55      3821
weighted avg       0.92      0.68      0.75      3821
```



Model Evaluation

Now we evaluate the performance of our Logistic Regression model using the test data.

Advanced Driver-Level Modeling

We will now aggregate the data per driver and create features that capture their performance trends.

```
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
```